A Real-Time Haptic Teleoperation Platform for UAVs Based on Unity–ROS Integration

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***Abstract*—Unmanned aerial vehicles (UAVs) are increasingly used in logistics, agriculture, inspection, and other complex tasks, yet effective human–robot interaction and real-time safety assurance remain challenging. In order to overcome the limitations of existing teleoperation platforms—such as visual feedback latency, complex and cluttered environments, and the absence of formal safety guarantees—this paper proposes a modular teleoperation platform that seamlessly integrates the Unity simulation environment, Robot Operating System (ROS), and a stylus-based Phantom (3D Systems Touch) haptic device. The platform features closed-loop, low-latency communication between operator, simulation, and computational modules, enabling natural 3D control and extensible safety logic such as control barrier functions (CBF). We demonstrate that combining real-time haptic feedback and CBF-based safety constraints significantly improves operator performance in a simulated corridor navigation task with static and dynamic obstacles. Compared to visual-only or single-modality control, our system reduces both task completion time and collision count, and provides intuitive feedback about proximity to obstacles. The results highlight the potential of this unified, extensible platform for advancing safe, user-friendly UAV teleoperation in challenging environments.**

***Index Terms*—Unmanned aerial vehicles, ROS, Haptic Feedback, Unity, Teleoperation.**

I. Introduction

The rapid development of unmanned aerial vehicles (UAVs) has enabled their widespread application in logistics, precision agriculture, industrial inspection, environmental monitoring, and search-and-rescue operations [1], [2], [3], [4]. For instance, Eun et al. showed that drone-based logistics can reduce CO_2 emissions compared to traditional vehicle delivery [5], while Benarbia and Kyamakya surveyed the feasibility and enabling technologies of drone delivery systems [6]. In emergency response, multi-UAV teams have been deployed for mapping and situational assessment [7], [8].

A representative use case is disaster response in dynamically changing flood zones, where multiple UAVs search for survivors and assess damage under tight time and safety constraints. In such missions, vehicles must navigate cluttered, partially flooded environments with moving debris and uncertain hazards under potentially degraded communications. Relying solely on visual feedback can be insufficient when latency and limited viewpoints degrade situational awareness. Prior work has demonstrated the effectiveness of multi-UAV strategies for survivor search in these settings [9], underscoring the need for operator interfaces that combine real-time safety and intuitive guidance.

To address environment dynamics at the multi-agent level, adaptive coverage and safe navigation methods have been developed. Bai et al. proposed adaptive multi-agent coverage with obstacle avoidance and later extended to safe adaptive coverage [10], [11], while Liu et al. studied safe 3D coverage for multi-agent systems [12]. Although such advances improve autonomy and robustness, their practical impact in teleoperation remains constrained by the lack of integrated platforms that provide real-time safety guarantees together with high-fidelity feedback and ease of use for human operators.

From a control standpoint, widely used UAV controllers originate from PID/cascaded baselines [13], [14], later augmented by adaptive and robust nonlinear methods [15]. However, these classical designs offer limited support for natural human interaction or on-the-fly mitigation of unforeseen obstacles in teleoperation.

A key challenge in practice is latency and limited perceptual bandwidth. Communication delay degrades the operator's situational awareness and can precipitate collisions or missed targets [16]. To mitigate cognitive load and improve guidance, haptic teleoperation has been explored: artificial force fields [17] and shared-autonomy/impedance approaches [18] provide repulsive or assistive cues near obstacles, complementing the visual channel and improving user performance in time-critical tasks [19], [20]. Yet, these methods typically lack formal safety guarantees. Control barrier functions (CBFs) address this gap by minimally modifying unsafe commands to keep the system within certified safe sets [21]; recent work has brought CBFs into haptic UAV teleoperation [22], and shown that appropriate haptic feedback can also improve user agreement and satisfaction in shared autonomy [23].

Another practical obstacle is fragmented toolchains. Many prior systems simulate vehicle physics and render haptics/visuals in separate stacks, incurring integration overhead and extra latency. Comparative studies report that ROS + Unity3D can offer higher visual realism and effective real-time sensor integration than traditional ROS–Gazebo pipelines [24], but UAV teleoperation platforms have yet to fully leverage a unified, high-fidelity framework.

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We present an integrated UAV teleoperation system coupling a stylus-based Phantom haptic device, Unity for high-fidelity visualization, and ROS as middleware. The operator's 3D stylus motions are mapped to UAV velocity commands, while a modular CBF layer enforces real-time safety and generates force feedback that intuitively guides the user away from hazards. This closed-loop design enables natural three-dimensional control while proactively preventing collisions and maintaining situational awareness. We validate the system in corridor navigation with static and dynamic obstacles, showing that haptic feedback significantly reduces collisions and task time compared to visual-only control. Our results indicate that a unified Unity–ROS–Phantom stack is a promising direction for safe and user-friendly UAV teleoperation.

II. Method

The primary goal of this work is to design an intuitive, safe, and extensible UAV teleoperation platform that enables natural human–robot interaction in complex 3D environments. Our system aims to support rapid operator input, high-fidelity simulation, low-latency bidirectional communication, and real-time safety assurance, while remaining modular for future extension to advanced control algorithms or alternative hardware.

A. System Architecture

To achieve these objectives, we construct a modular system architecture (see Fig. 1) that tightly integrates three core components:

The proposed platform consists of three tightly-coupled modules, each responsible for a distinct aspect of the teleoperation pipeline. First, the Haptic Device Module leverages a stylus-based Phantom device to capture the operator's hand movements in all three spatial dimensions. Position data from the device are read via USB at high frequency, ensuring low-latency and high-precision input for teleoperation. Unity interfaces directly with the Phantom device, converting stylus motion into UAV velocity control commands, which are then passed to the simulation loop and published over the communication backbone.

Fig. 1. Software architecture: the Phantom haptic device, Unity simulation, and ROS computation modules form a real-time closed loop, connected via USB and TCP/IP.

Second, the Simulation and Visualization Module is realized in Unity, which not only provides a photorealistic 3D environment—including UAV kinematics, static and dynamic obstacles, and environmental interactions—but also acts as the primary visual interface for the operator. Within this module, UAV states, obstacle proximities, and environmental information are continuously updated, and all relevant data are exchanged with ROS in real time using the TCP/IP protocol. Unity also receives safety-filtered commands and haptic feedback forces from ROS, applying them to update the simulation and drive the force rendering engine.

Third, the Communication and Computation Module employs ROS2 as the core middleware. This module subscribes to all necessary state and sensor topics published by Unity, performs centralized processing (including real-time safety enforcement via CBF and QP solvers, as well as haptic force computation based on obstacle proximity and CBF constraint violations), and publishes updated UAV control commands and haptic feedback force values back to Unity for execution. The use of TCP/IP ensures robust, low-latency inter-process communication between Unity and ROS, even when running on separate operating systems or containers.

This layered, modular structure enables flexible development and extension of each subsystem, while the unified data flow closes the loop between user input, simulation, control logic, and feedback. As a result, the system achieves both real-time performance and high fidelity in operator experience, safety assurance, and teleoperation effectiveness.

B. Control Logic and Safety Layer

The basic operation logic of the proposed system is straightforward. The Phantom device records the operator's three-dimensional stylus motion, which is mapped directly to the UAV's velocity command in Unity. The operator control mapping is defined as

$$\mathbf{v} = k\mathbf{P}_{Phantom} \tag{1}$$

where $\mathbf{v}$ is the UAV velocity command, $\mathbf{P}_{Phantom}$ is the Phantom's current position vector (relative to its neutral pose), and k is a tunable proportional gain. This direct mapping enables intuitive, real-time UAV teleoperation similar to manipulating a joystick or game controller, but with richer three-dimensional input.

To guarantee safe navigation in environments with obstacles, a CBF is integrated as the safety core of the system. The CBF encodes a safety constraint based on the UAV's distance to nearby obstacles:

$$h(\mathbf{x}) = \|\mathbf{x} - \mathbf{x}_{\text{obs}}\|^2 - r_{safe}^2 \tag{2}$$

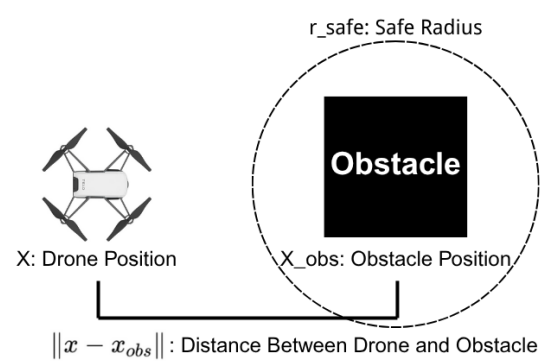

Fig. 2. CBF in the system

where $\mathbf{x}$ is the UAV position, $\mathbf{x}_{\text{obs}}$ is the position of the nearest obstacle, and r_{safe} is the minimum required safety radius.

Safety-constrained velocity control commands $\mathbf{u}$ are computed by solving a quadratic program (QP) that minimally adjusts the operator's input $\mathbf{v}$ to ensure $h(\mathbf{x}) \geq 0$ is satisfied:

$$\min_{\mathbf{u}} \|\mathbf{u} - \mathbf{v}\|^2 \tag{3}$$

$$\text{subject to: } 2(\mathbf{x} - \mathbf{x}_{\text{obs}})^{\mathsf{T}}\mathbf{u} + \gamma h(\mathbf{x}) \geq 0 \tag{4}$$

where γ is a positive constant for the CBF. This guarantees the UAV remains in the safe region, while only minimally modifying the operator's desired command.

C. Safety-Enhanced Architecture with CBF Module

The architecture with the CBF safety module is illustrated in Fig. 3. Compared to the base system, the ROS computation module now includes a real-time safety layer. All sensor data, control commands, and obstacle information are routed through the CBF/QP logic, which filters or adjusts unsafe commands before they are sent to Unity for execution. At the same time, the difference between the raw and safe command is used to compute haptic feedback signals, forming a closed-loop pipeline for both control and operator guidance.

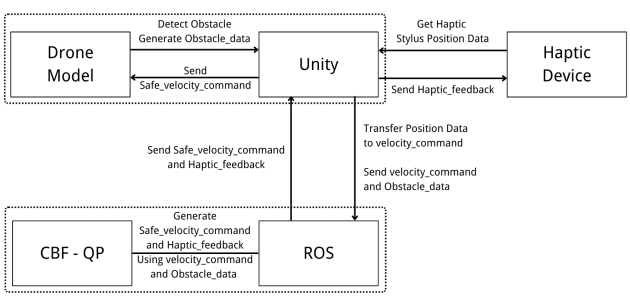

Fig. 3. System architecture with integrated safety module (CBF)

D. Haptic Feedback Generation

To support intuitive and immersive teleoperation, the system delivers real-time haptic feedback through the Phantom device. The feedback force is a smooth, continuous function of the safety margin or the magnitude of CBF-induced command modification. Specifically, the force increases as the UAV nears an obstacle or as the operator's input is more heavily constrained by the CBF. The direction of the force pushes the operator away from unsafe regions.

To maintain natural interaction, when obstacles suddenly appear or disappear (e.g., a moving obstacle leaves the UAV's vicinity), the feedback force is gradually ramped up or down using a decay function, rather than changing abruptly. This design prevents discomfort and helps the operator smoothly adapt their strategy.

Overall, this modular, safety-enhanced architecture enables real-time, closed-loop teleoperation, intuitive operator feedback, and straightforward extensibility for advanced algorithms or new hardware components.

III. Experimental Setup

The main objectives of the experimental study are to evaluate the feasibility, stability, and usability of the proposed teleoperation platform. Specifically, we aim to verify that the system enables intuitive and precise UAV control, supports seamless integration of safety modules (such as CBF), and delivers effective haptic feedback to the operator.

A. Hardware and Runtime Environment

All experiments were performed on a single workstation running Windows 11, equipped with an Intel Core i9-12900K CPU, 32 GB RAM, and an NVIDIA GeForce RTX 3090 GPU. The Unity simulation environment (version 2020.3) and Phantom haptic device drivers operated natively on Windows 11, while the ROS2 computation module was deployed in a Docker container using the tiryoh/ros2-desktop-vnc:humble image. Communication between Unity and ROS2 was conducted over the local TCP/IP interface (127.0.0.1). All components were executed on the same physical host, minimizing network latency and ensuring optimal conditions for teleoperation and feedback evaluation.

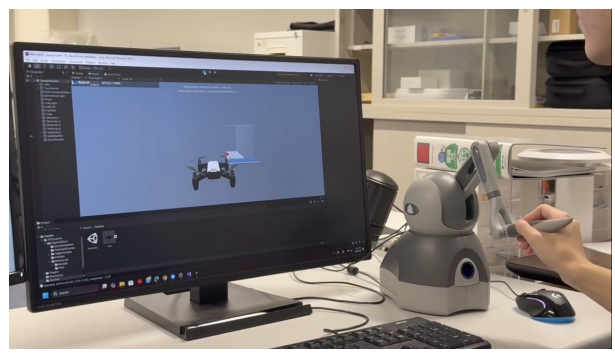

Fig. 4. Experiment environment.

B. Scenario Setup

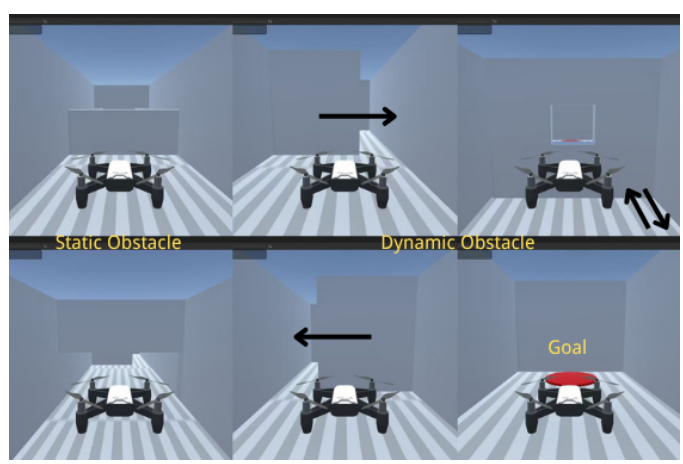

Fig. 5. Obstacles in the Unity corridor environment.

The simulated environment consists of a straight corridor containing both static and dynamic obstacles. Static obstacles

are fixed along the UAV's flight path, while dynamic obstacles move along predefined trajectories—such as sliding sideways, forward and backward. This configuration requires the operator to not only plan safe routes in advance, but also adapt to obstacles that may unexpectedly block the UAV's path.

C. Experimental Procedure

Five voluntary participants, each with at least five minutes of training using the Phantom haptic device, were recruited for the study. Each participant performed the teleoperation task under four experimental conditions presented in sequence: (1) a baseline mode with neither CBF nor haptic feedback, relying solely on visual information; (2) a CBF-only mode, in which the safety layer was enabled but no haptic feedback was provided; (3) a haptic-only mode, where force feedback was delivered but without CBF safety constraints; and (4) a combined CBF and haptic mode, enabling both safety constraints and force feedback simultaneously.

For each condition, the participant teleoperated the UAV from the corridor entrance to a designated landing zone, aiming to avoid all obstacles. Throughout the experiment, all obstacle positions—both static and dynamic—were detected by the UAV using Unity's raycasting mechanism and continuously published to ROS in real time.

In the experimental setup, the CBF and haptic feedback modules operated independently, allowing their individual and combined effects to be evaluated. In the "CBF Only" condition, the CBF module served solely as a safety mechanism: operator commands were automatically adjusted to keep the UAV outside a predefined safe radius of obstacles, thus preventing collisions. However, no force feedback was provided, even if the UAV approached obstacles closely. Conversely, in the "Haptic Only" condition, force feedback was generated based purely on the UAV's real-time distance to obstacles, giving the operator intuitive cues as hazards approached, but without enforcing any formal safety margin—so collisions could still occur. When both modules were enabled ("CBF + Haptic"), haptic feedback was further refined by incorporating the safe radius defined by the CBF, making the force cues more closely aligned with the safety boundary and enabling proactive operator adjustment as constraints were enforced.

This design allowed the study to independently and jointly assess the effects of formal safety enforcement and intuitive haptic feedback on operator performance and overall system safety.

D. Performance Metrics and Data Collection

To comprehensively evaluate the effectiveness and usability of the proposed teleoperation platform, we collected detailed data across multiple dimensions during each trial. Specifically, the following performance metrics were recorded:

We evaluated three key performance metrics: (1) task completion time, defined as the total time each operator took to fly the UAV from start to landing zone under each condition, reflecting control efficiency and intuitiveness; (2) collision count, i.e., the number of contacts between the UAV and any obstacle per trial, indicating overall safety performance; and (3) minimal obstacle distance at reaction, the closest distance between UAV and obstacle at the instant the operator first applied a corrective maneuver, which measures how effectively the feedback modalities alerted the user to impending collisions.

These metrics were recorded via ROS topics, with UAV states, obstacle proximities, operator inputs, and haptic feedback forces continuously timestamped and logged for analysis.

Additionally, participants completed a post-experiment questionnaire assessing five subjective dimensions—Ease of Use, Safety, Intuitiveness, Immersion, and Fatigue—on a 5-point Likert scale, providing qualitative insights to complement quantitative results.

IV. Results

Experimental results highlight the effectiveness of the proposed teleoperation system in improving operator performance and ensuring UAV safety across all test conditions.

A. Task Completion Time

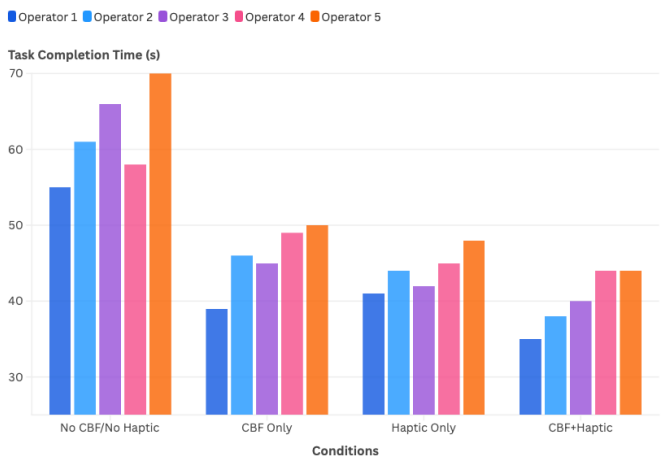

Fig. 6. Task completion times under different conditions for each operator.

TABLE I
Task Completion Time (s) for Each Operator under Different Conditions

Condition	Op1	Op2	Op3	Op4	Op5
No CBF/No Haptic	55	61	66	58	70
CBF Only	39	46	45	49	50
Haptic Only	41	44	42	43	48
CBF+Haptic	35	38	40	44	44

Fig. 6 shows the task completion times for five operators under different experimental conditions. When neither CBF nor haptic feedback was enabled, completion times were the highest and most variable across operators. Enabling either CBF or haptic feedback individually reduced both average time and variability. The combination of CBF and haptic feedback yielded the shortest and most consistent completion times for all operators, demonstrating improved efficiency and stability in navigation.

B. Collision Count

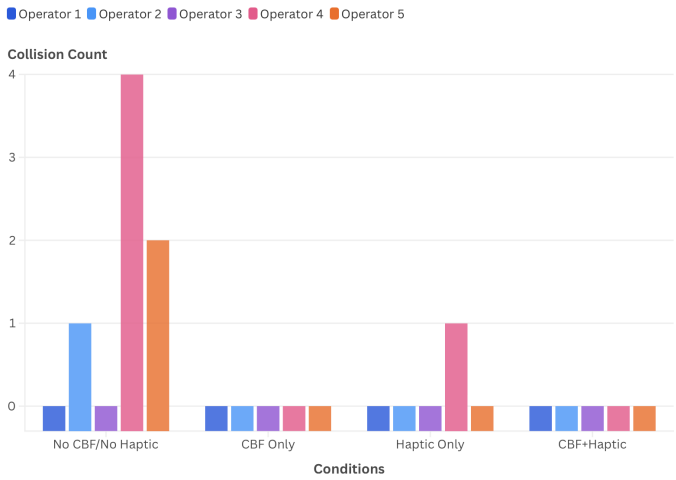

Fig. 7. Number of collisions under each condition for all operators.

The collision count results, shown in Fig. 7, further support the advantage of the proposed system. Operators experienced multiple collisions in the absence of both CBF and haptic feedback. Both interventions alone reduced collisions, while the combined approach effectively eliminated collisions for all operators, highlighting the synergistic effect of integrating CBF with real-time force feedback. The detailed collision count data for each operator under different experimental conditions are listed in Table II.

TABLE II
COLLISION COUNT FOR EACH OPERATOR UNDER DIFFERENT CONDITIONS

Condition	Op1	Op2	Op3	Op4	Op5
No CBF/No Haptic	0	1	0	4	2
CBF Only	0	0	0	0	0
Haptic Only	0	0	0	1	0
CBF+Haptic	0	0	0	0	0

C. Haptic Force and Obstacle Proximity

To further understand how haptic feedback assists operators in perceiving and responding to obstacles, Fig. 8 plots the relationship between the UAV's distance from obstacles and the corresponding force rendered by the Phantom device. As the UAV approaches obstacles, the haptic feedback force increases smoothly, providing timely cues for the operator to adjust their path. This continuous and proportional force feedback mechanism enables operators to react before breaching the safety margin, leading to improved safety and fewer last-moment evasive maneuvers.

It should be noted that in Figure 8, the force values $F(t)$ appear smooth because they were sampled at 0.5-second intervals during the experiment. This relatively low sampling rate was intentionally chosen to make the overall trend between force and distance more visible and to ensure the experiment proceeded smoothly, rather than to capture the high-frequency fluctuations typical of haptic force signals. No additional filtering or post-processing was applied.

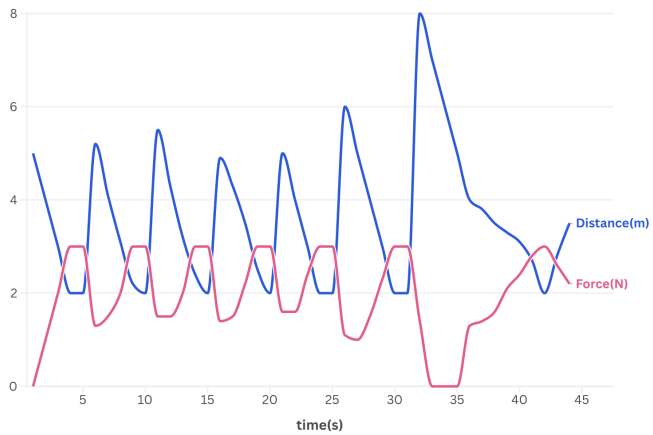

Fig. 8. Relationship between distance to obstacles and haptic feedback force for operator 3 during the CBF and Haptic enabled trial.

D. User Experience Evaluation

To further assess subjective user experience, all participants completed a post-experiment questionnaire after each experimental condition, rating the system on five dimensions: Ease of Use, Safety, Intuitiveness, Immersion, and Fatigue. Ratings were given on a 5-point Likert scale (higher scores indicate better performance, except for Fatigue where higher is less fatigue). The summary of results is shown in Fig. 9, and the mean scores for each experimental condition are detailed in Table III.

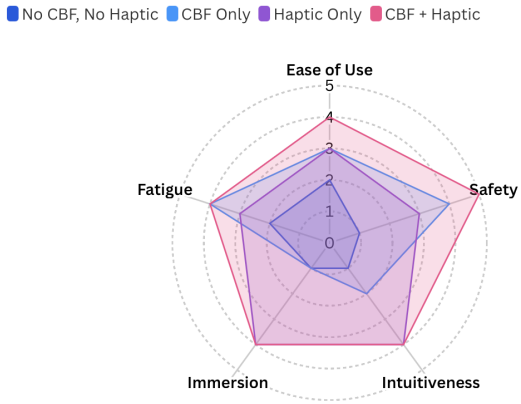

Fig. 9. Radar chart of subjective user ratings across five evaluation criteria under each system condition.

TABLE III
SUBJECTIVE QUESTIONNAIRE RESULTS (MEAN SCORES FOR EACH CONDITION)

Condition	Ease	Safety	Intuit.	Imm.	Fatigue
No CBF, Haptic	2	1	1	1	2
CBF Only	3	4	2	1	4
Haptic Only	3	3	4	4	3
CBF + Haptic	4	5	4	4	4

As shown in Fig. 9, the CBF + Haptic condition received the highest scores across all metrics. Participants reported that this setting was the safest and most intuitive, with the greatest

sense of immersion and the lowest levels of fatigue. Both CBF Only and Haptic Only conditions also outperformed the baseline (No CBF, No Haptic), but their improvements were not as pronounced as the combined mode. The baseline system scored lowest in all dimensions, with particular deficits in safety and immersion.

These subjective results are consistent with the objective performance metrics, confirming that integrating CBF-based safety constraints with real-time haptic feedback not only enhances operator safety and efficiency, but also significantly improves overall user experience.

E. Discussion

Collectively, these results demonstrate that integrating both CBF and haptic feedback leads to more reliable, safer, and faster UAV teleoperation compared to conventional visual-only or single-modality approaches. The haptic force feedback, in particular, improves operators' awareness of nearby obstacles, facilitating proactive corrections and reducing both collision risk and task completion time. These findings confirm the practical value of the proposed platform for real-world UAV teleoperation in complex, dynamic environments.

V. Conclusion

This paper proposed a modular, extensible teleoperation platform that integrates a stylus-based Phantom haptic device, Unity-based 3D simulation, and ROS middleware for real-time UAV control. The system features seamless, low-latency bidirectional communication and supports the integration of advanced safety algorithms such as CBF. Experimental results in a realistic, obstacle-rich simulated environment demonstrated that the combination of CBF and haptic feedback significantly reduces both task completion time and collision risk compared to baseline visual-only or single-modality approaches. The closed-loop haptic feedback mechanism provides operators with intuitive cues about proximity to obstacles, facilitating safer and more efficient UAV navigation.

While our integrated Unity–ROS–Phantom platform demonstrates promising results, several limitations remain. All experiments were conducted in a virtual corridor, and real-world scenarios may introduce additional challenges such as sensor noise, communication packet loss, and more unpredictable dynamics. The user study also involved a limited number of participants. Furthermore, only CBF-based safety logic was evaluated; the framework, however, is modular and can accommodate advanced safety and shared autonomy modules.

Future research will focus on expanding experimental scenarios to include multi-UAV, outdoor, and VR-based immersive teleoperation, increasing the number of participants, and integrating machine learning for adaptive feedback and user modeling. We believe this platform lays a strong foundation for robust, safe, and user-friendly UAV teleoperation in dynamic, real-world applications.

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